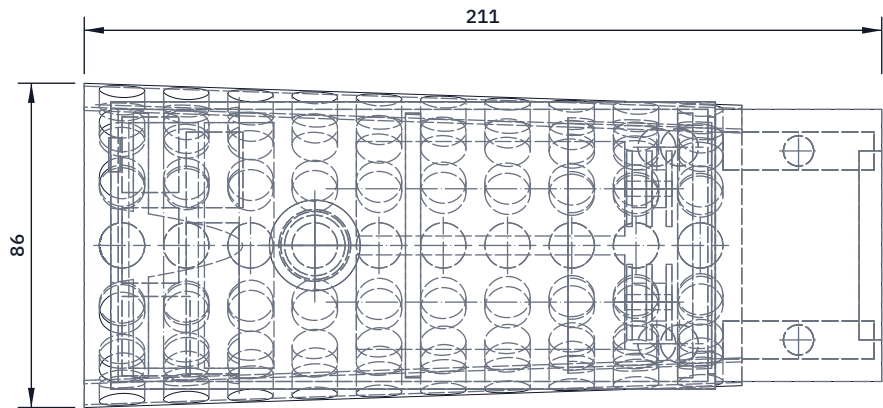


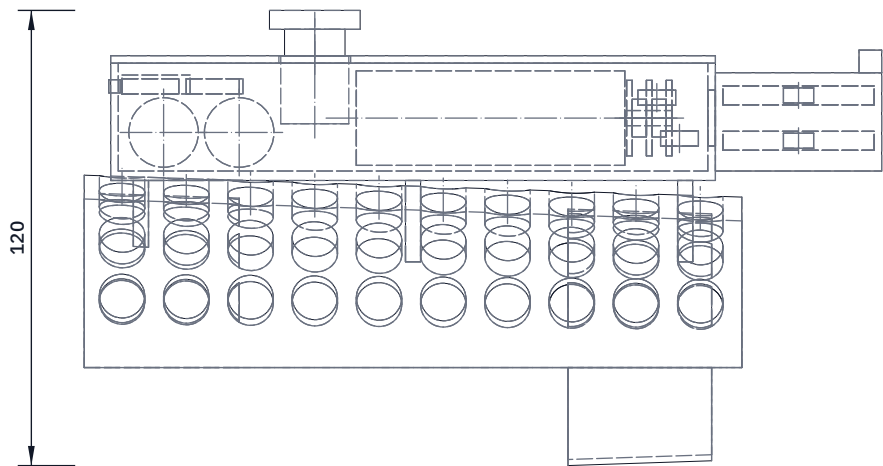
CONCEPT, NOT FOR FABRICATION

FlexHand · github.com/BoujeeEnjinia1701/flexhand



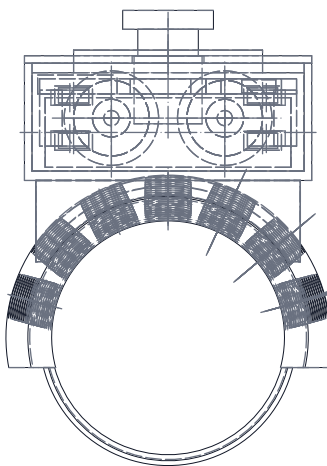
TOP VIEW

Scale 1:2; looking down (along -Z)



FRONT VIEW

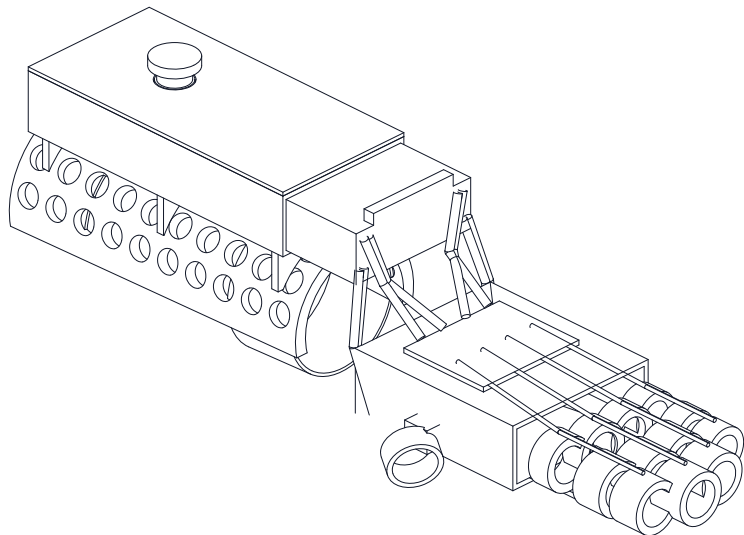
Scale 1:2; looking at the front (along +Y)



RIGHT VIEW

Scale 1:2; looking at the right side (along -X)

REV	DESCRIPTION	DATE	BY
P1	General arrangement for TRL 3 (FXH-CAL-001)	2026-09-25	AC
P2	FXH-DDR-002: balance pulleys, 44 mm anchor block, perforated cuff, thimbles	2026-09-25	AC



ISOMETRIC VIEW, WHOLE DEVICE

Left hand shown (mirror of the modeled right hand); not to scale; seen from the front right and above, about 30 deg elevation

MAIN DIMENSIONS (MM)

Pack 160 long x 76 wide; walls 2, lid 2
Pack top 74.5 above forearm axis; wrist to pack front 55
Gearmotors 25 dia x 71, axes along forearm at y = +/-15
Spools: core 10 dia, flanges 20 dia, 12 wide
Tendon exits at y = +/-31; idlers 10 dia (623ZZ class)
Cells 2 x 18650 across the pack at x = -201, -181
Forearm cuff 174 long; shell 2, liner 4; 12 dia holes; straps 38 wide
Anchor block 44 long; balance pulleys 8 dia at y = +/-25, 32 travel
Anchor cuffs (index, middle, ring, little): 16.4, 18.6, 17.0, 12.2
Thimbles (index, middle, ring, little): 17.0, 19.0, 17.5, 13.2
Guide cuffs 12 wide; plates TPU 2.5 thick

PARTS LIST (ITEMS MATCH BOM/BOM.CSV)

- Forearm cuff with liner and straps
- Pack base
- Gearmotor with encoder (2)
- Antagonistic spool (2)
- Li-ion cells and 2S BMS
- Controller
- Motor driver (2)
- Pack lid
- Emergency stop
- Sheath anchor block

PARTS LIST, CONTINUED

- Bowden sheaths (8)
- to 17, 21 Hand side (iso only)
- USB-C charger
- Wiring (not shown)
- Idler bearings (4)
- Balance pulleys (4)
- Not a medical device.
- Research use under supervision.
- Never charge while worn.

FLEXHAND

github.com/BoujeeEnjinia1701/flexhand

Forearm unit general arrangement

DRAWING NO. FXH-DWG-001		REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:2	UNITS mm	PROJECTION 	DATE 2026-09-25	DRAWN Amish Chadha
MATERIAL / NOTES Pack and cuff PETG; plates and cuffs TPU 95A; bought parts per bom/bom.csv. PRELIMINARY, NOT FOR FABRICATION				
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